ALEXANDER MORTILLITE

Software Developer • Air Traffic Systems • AI-Assisted Development
amortillite@gmail.com • (609) 457-6508 • Egg Harbor Twp, NJ • GitHub • LinkedIn • Portfolio
🔒 **Active Public Trust Security Clearance**

PROFESSIONAL SUMMARY

Recent Computer Science graduate with hands-on experience in efficiency-critical aviation software development at the FAA's William J. Hughes Technical Center through Enroute Computer Solutions and former internships. Holds an active Public Trust security clearance. Skilled in C++, Python, and Java with practical experience applying AI-assisted development tools (Claude, ChatGPT) to accelerate coding, debugging, and documentation workflows. Adept at validating and refining AI-generated outputs, with a track record of delivering modular, well-documented and maintainable code in real-world software environments.

TECHNICAL SKILLS

Languages: C++, Python, Java, JavaScript, HTML/CSS
AI Tools: Claude, ChatGPT - prompt engineering, code validation, AI-generated output review
Frameworks & Libraries: Qt (familiar), Raylib, Jakarta EE, LibGDX, THREE.js and Tailwind CSS.
DevOps & Systems: Git, CI/CD pipelines, Linux, Containers, Virtual Machines, VPNs, Microcontrollers
Data & Databases: SharePoint, Power Automate, PowerApps, JSON, CSV data processing
Core CS: OOP, Data Structures & Algorithms, Software Engineering, REST APIs, MVC architecture

EDUCATION

B.S. Computer Science | Minor: Business Administration — *GPA 3.4*
Stockton University, Galloway, NJ — Graduated May 2025

Academy of Computer Science | Atlantic County Institute of Technology — *GPA 3.6*
Mays Landing, NJ — Graduated May 2021

Relevant Coursework: Artificial Intelligence · Software Development · Data Structures & Algorithms · Data Mining · Probability & Applied Statistics · Mobile App Development · Information Assurance · Calculus I & II

PROFESSIONAL EXPERIENCE

Software Engineer / DevOps Automation / System Administration

Enroute Computer Systems (EOSS Contractor) — FAA William J. Hughes Technical Center — *December 2025 – Present*
Enroute Air Traffic Management Systems: DOTS+, MEARTS, BUILDS & CONTROLS

- Developed and prototyped a web application of the DOTS+ system in THREE.js, HTML, and CSS.
- Designed and produced powerful libraries for rendering/vector functions with built in redundancy.
- Wrote detailed documentation using in-file JSDoc for continued maintenance.
- Implemented a Python script converting fixed airspace position information (.dat) to a modern JSON format.
- Utilized Object Oriented principles to create clear and impactful data structures.
- Managed branching and Git repositories for codebases with over 4,000+ lines.
- Reconfigured a DOTS+ build box for the assessment of security vulnerabilities.
- Worked with stakeholders to develop navigable CI/CD Build Automation tools for Builds & Controls (B&C).
- Successfully built and distributed ERIDS SSM-116 with the B&C Team.
- Supported the MEARTS Lab Lift through physical management and installation of network infrastructure.

Software Developer / Project Lead (CS Intern)

Federal Aviation Administration — AJM-2512 (Gateways Program) — *June 2024 – January 2025*
Enroute Air Traffic Management Organization: SPECIALTY ENGINEERING

- Developed applications and automations which facilitated high-speed, executive information exchange.
- Led meetings with stakeholders to establish feature expectations and provide real-time updates.
- Created and maintained clear, Evergreen Documentation, utilizable in further application development.
- Streamlined Database Report Generations (DRGs) through Power Automate and SharePoint List.
- Practiced full-stack development in a real-world software development environment.
- Used scripting languages like HTML and CSS to create automated, stylized PDFs.
- Utilized Microsoft PowerApps to create seamless integrations of databases into Microsoft Teams.
- Created and managed multiple SharePoint databases, supporting cross-functional communication.

RELATED EXPERIENCE

Programmer / Project Lead
FTC Robotics — David Wood 4-H Center *September 2015 – December 2020*

- Established libraries used in the production of autonomous and user-controlled robots.
- Conducted real-time testing and troubleshooting of robots within dynamic three-dimensional environments.
- Deployed critical software updates between competition rounds to address emerging issues.
- Taught junior team members fundamentals of Java and Android Studio.
- Communicated concurrently with build teams to prevent critical failures before they happened.
- In tandem with building teams, tested and evaluated different variations of tracked vs. wheeled motion.

SELECTED PROJECTS

Portable CO2 Air Quality Monitor — *C++ · ESP32 · Node.js · Chart.js · Caddy* *January 2026 – Present*

- Hypothesized elevated CO2 levels in my room were degrading sleep quality; I built a portable ESP32 + SCD41 sensor device to test it, including soldering hardware components onto a breadboard.
- Wrote a C++ driver in Arduino IDE to read CO2, humidity, and temperature; transmitted data via secured POST requests (128-bit key + device whitelist) to a self-hosted Node.js server on an Intel NUC running Fedora Server.
- Configured a Caddy reverse proxy to expose the server over HTTPS, built a Chart.js WebUI to display live 24-hour sensor buffers, and implemented auto-recovery logic to reload stored data on server restart.

Laser Reflection Simulation — *C++ · Raylib · 2D Physics* *Summer 2025*

- Developed a ray marching algorithm in 2D space to simulate real-time laser reflection and collision, computing surface normals to derive dynamic incidence angles and reflection vectors.
- Built interactive controls for circle spawning, laser direction, pause/rotation modes, and single-object removal; leveraged prior unit vector experience from the planetary simulator to accelerate development.

Quadtree Division System — *C++ · Raylib* *Spring 2025*

- Built a reusable Quadtree library for efficient 2D spatial partitioning; applied it to a Barnes-Hut orbital simulation, reducing complexity from $O(n^2)$ to $O(n \log n)$.
- Authored comprehensive test cases to verify data structure accuracy and edge-case correctness.

Advanced Solar Simulator — *Java · LibGDX · Physics* *Winter 2024*

- Simulated planetary motion with physics-accurate thrust vectoring using trigonometry; implemented real-time trajectory projection to visualize collision forecasting.

Knowledge Discovery & Data Mining — Research Project — *Python · Data Analysis* *Fall 2023*

- Cleaned and analyzed large multi-state survey datasets using Python; uncovered correlations between Alzheimer's and lifestyle factors through custom algorithm design.
- Led a team of four, managing planning, task delegation, and version control via GitHub.

Probability & Statistics Library — *Java · Swing GUI* *Spring 2023*

- Built a suite of statistical algorithm solvers covering Hypergeometric, Poisson, and Chebyshev distributions; implemented a data salting and smoothing pipeline from scratch with custom methods alongside Apache Commons Math comparisons.
- Developed a full MVC GUI application with JFreeChart visualizations; conducted an applied research analysis on Age of Empires 2 matchup data from Kaggle to evaluate game balance using statistical inference.

CERTIFICATIONS

MTA: Networking Fundamentals (2021) • MTA: Windows OS Fundamentals (2019) • IC3: Digital Literacy (2018)

PROFESSIONAL REFERENCES

Chris Carboy — AJM-25 Software Development Manager, FAA • (609) 485-8288
Derek Kubont — AJM-2512 Specialty Engineering Manager, FAA • (609) 233-3600
Kevin Tracy — AJM-25 Security Team Lead, FAA • Kevin.R.Tracy@faa.gov